OpenRemote

URL: https://openremote.io/

OpenRemote is an intuitive, user-friendly, 100 per cent open source IoT device management platform for original equipment manufacturers (OEMs), integrators and governments. Its functions and features include building applications for services, configuring dashboards; building logic with When-Then, Flows, Javascript or Groovy; and customised apps under the AGPLv3 license. The major domains it is used in include energy management and smart cities.

Mainflux

URL: https://mainflux.com/

Mainflux is a performant and secure open source IoT platform with full-scale capabilities for developing IoT solutions, IoT applications and smart connected products. Built as a set of microservices containerised by Docker and orchestrated with Kubernetes, the Mainflux IoT platform serves as a software infrastructure and middleware which provides device management, data aggregation and data management, connectivity and message routing, core analytics, and application enablement. Mainflux Labs (under an Apache 2.0 license) is the technology company that offers end-to-end, open source, edge computing gateway and consulting services for the software and hardware layers of IoT technology.

ThingsBoard

URL: https://thingsboard.io/

This platform offers the Community Edition, the Professional Edition and Cloud. The Community Edition is completely free. ThingsBoard is an open source IoT platform meant for device management, data collection, processing and visualisation to build the IoT solution. This enables you to connect devices, push data from devices, build real-time end-user dashboards, define thresholds, trigger alarms, and push notifications about new alarms over email, SMS or other systems.

Thinger.io

URL: https://thinger.io/

In Thinger.io you have an easy-to-navigate, ready-to-use cloud infrastructure that allows users to integrate, monitor, and control millions of IoT devices. It helps to connect an API, supports geofencing, and can create custom alerts using real-time data from devices. It permits you to collect and store data in the platform, and create dashboards with custom or built-in widgets. There is also a provision to share dashboards with customers and on the cloud.

Thinger.io offers the following three choices – Community, Private Cloud and On-Premise. The Community plan provides free access for evaluation and testing that is limited to two devices, simple analytics and a single tenant.

Kaa

URL: https://www.kaaiot.com/

Kaa provides a flexible, multi-purpose middleware platform for developing end-to-end IoT solutions, connected applications, and smart devices. Its products are diversified into Kaa IoT Platform, Kaa Cloud, Kaa IoT Gateway, Kaa IAM and Kaa Cloud Migration. It has applications in the domains of agriculture, healthcare, automation, Industrial IoT, wearables and logistics. It offers an end-to-end IoT development experience with its customisable microservice architecture and open protocols.

Table 1: Features of the various IoT platforms

IoT platform	Integration	Protocols	Security	Free and open source
Arduino IoT Cloud	REST APIs	MQTT	Basic authentication	Up to two devices
OpenRemote	WebSocket, HTTP, MQTT APIs	HTTP, MQTT, KNX	AES, 256-bit encryption	Yes
Mainflux	REST APIs	HTTP, MQTT, CoAP	TLS v1.3	Yes
ThingsBoard	WebUI, REST APIs	HTTP, MQTT, CoAP	Basic authentication	Yes
Thinger.io	REST APIs	HTTP, MQTT, CoAP	SSL/TLS	Up to two devices
Kaa	Portable SDKs	HTTP, MQTT, CoAP	TLS / DTLS	Up to five devices
Blynk	REST APIs	HTTP, MQTT	Basic authentication	Up to two devices
SiteWhere	REST APIs	MQTT, AMQP	SSL	Yes
ThingSpeak	REST APIs, MQTT APIs	MQTT	Basic authentication	Yes
Zetta	WebSocket, REST APIs	MQTT	Basic authentication	Yes